

The Engel correction cannot tend to zero

Disproof of Conjecture 00000002760

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Scope. The problem proposes that the dimension of k -Engel linear matrix spaces is $n^2 - n + c(k)$, where $c(k)$ decreases to zero as k grows. Neither language version excludes $n = 1$. We give a counterexample in this matrix size, which also works if the stated dimension is interpreted as the maximum possible dimension.

Actual iterated commutators. Let $M = M_1(\mathbb{Q})$ and take the full linear space $S = M$. For matrices $X, Y \in M$, define

$$E_0(X, Y) = Y, \quad E_{k+1}(X, Y) = [X, E_k(X, Y)] = XE_k(X, Y) - E_k(X, Y)X.$$

These are the usual iterated Engel commutators. Every 1×1 matrix is a scalar matrix, so actual matrix multiplication commutes. It follows immediately that $E_k(X, Y) = 0$ for every $k \geq 1$ and all $X, Y \in M$. Thus the full space is k -Engel for every positive k . The same conclusion holds if the repeated commutator is conventionally taken on the other side.

Dimension and maximality. The matrix (1) is a basis of M over $\mathbb{Q}$, so $\dim_{\mathbb{Q}} S = 1$. Every linear subspace of M has dimension at most one. Consequently the maximum dimension of a k -Engel linear space in $M_1(\mathbb{Q})$ is also exactly one for every positive k .

Contradiction. Apply the proposed formula at $n = 1$. Since the actual dimension is one, it forces

$$1 = 1^2 - 1 + c(k) = c(k) \quad \text{for every } k \geq 1.$$

Hence the correction has a constant tail equal to one and converges to one. It cannot converge to zero: uniqueness of limits would give $1 = 0$. Monotonicity does not remove this contradiction. Therefore no correction function with the stipulated properties can satisfy the claimed dimension formula, and the conjecture is false.

Formal certificate. The Lean project uses the actual type of 1×1 rational matrices, its genuine multiplication, the recursive iterated commutator, and the full matrix submodule. It proves the Engel property for every positive index, calculates its finrank, bounds every subspace's finrank and certifies the maximal dimension. The final theorem rules out any real-valued correction satisfying the formula, monotonicity and convergence to zero. Printed axiom audits use only the standard axioms of Lean and Mathlib.